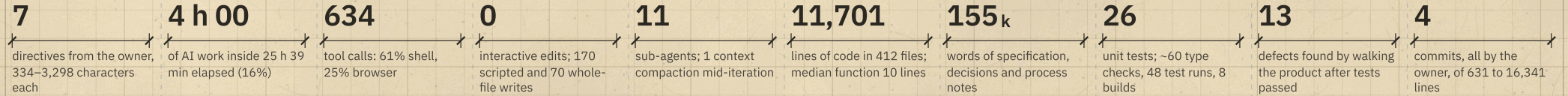


River Portal × Claude Opus 5.5

How one AI model specified, built, verified and documented a bilingual charity portal for one human owner, from an empty folder to a public site and a staff studio, on 24–25 September 2026.

§ 01 MEASURED — GIT, CODE, SESSION LOG. SELF-OBSERVATION: READ WITH CARE



§ 02 THE TASK

River Portal: a charity portal where needs meet gifts, carried by coordinators and carriers. Privacy is strict, a gift is a gift, and every public number comes from an append-only log.

«...The flows carry what is needed to those who need it»

THE OWNER'S FIRST DIRECTIVE: BANKS AND OUTFLOWS CARRY WHAT IS NEEDED TO THOSE WHO NEED IT

Delivered in iterations 00–04: a 146-note specification, 23 ADRs, 17 packages behind gates, a public site and a studio, a Pulumi program, 9 demo personas and 3 organisation profiles.

NEXT.JS 16 · REACT 19 · TYPESCRIPT 5.9 · FIRESTORE · VERTEX AI · PULUMI ON GOOGLE CLOUD · EN-GB + UK

FIG. 2 — WHERE THE 13 DEFECTS OF ITERATION 04 SURFACED



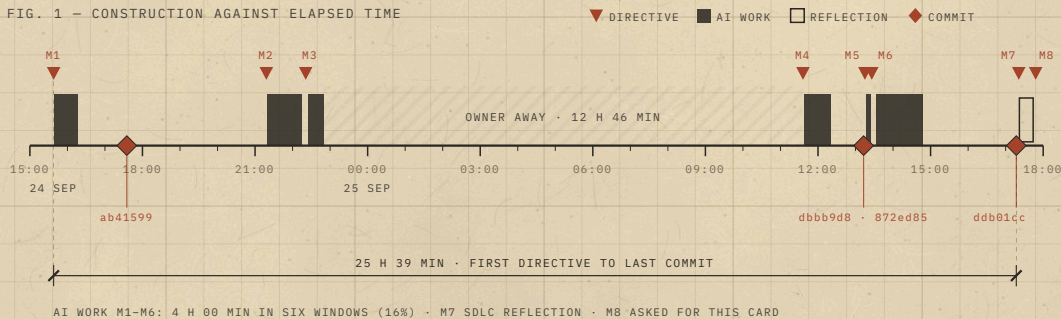
§ 03 HOW THE MODEL GENERATED CODE

- Bottom-up by layer.** Foundation, record, domain, surface, compose, app. After each change lower down, the type checker's error list became the work list.
- Wide, guarded edits.** Not one interactive edit in the main thread. Each of 170 scripted replacements asserted a single match, alongside 70 whole-file writes. Fast across many files, and loud when the code was not as expected.
- Gates as context.** It planned from 17 gate files, not 285 sources. One exported function per file kept units small: a median of 10 lines.
- A check for every 1.5 writes.** Type checks, test runs, topology checks, production builds and 30 screenshot runs, interleaved with building.
- Text in parallel, code in one thread.** Sub-agents wrote the specification and 1,029 lines of bilingual content. Parallel text drifted in numbers, names and privacy, and needed harmonising.
- Decisions written as built.** ADRs followed verified code and record what testing found. The owner saw most of them only afterwards.
- Defects at the boundaries, not in the logic:** framework serialisation, the CSS cascade, Ukrainian inflection, scenario continuity, the meaning of a parameter.
- Amplification.** A 472-character directive became 263 changed files. Acceptance rested on evidence (screenshots, check output, evaluations), not on reading diffs.

§ 04 IN THE MODEL'S OWN WORDS

- “ I am fastest where rules are mechanical: types, topology, gates. I am slowest where meaning is involved: what a number says to a donor, or what a label reveals about a village.
- “ The browser taught me more than my tests. With everything green, walking each role still found 13 defects.
- “ Two flaws were my own designs: settings that became ‘custom’ when saved unchanged, and a demo that could not be walked. Using the product found them; reading the code did not.
- “ The vaults are my memory. After my context was compacted mid-iteration, I carried on from files, not from recollection.
- “ I reviewed my own work. That is the weakest link: a fresh reviewer would see what I cannot.
- “ Construction took 16% of the elapsed time. The owner's attention set the pace, so easy acceptance matters more than faster building.

FIG. 1 — CONSTRUCTION AGAINST ELAPSED TIME



§ 05 NEXT TIME

- ☐ Restate each directive before building: goal, scope, evidence.
- ☐ Commit by layer, citing the ADR and the iteration.
- ☐ Run the walk, the crawl and the screenshots in CI.
- ☐ Add an independent review before acceptance.

In full: reflections/sdlc/v1, folder 10-proposals.

MODEL	Claude Opus 5.5 · claude-opus-5-5, all 1,407 replies
HARNESS	Claude Code desktop: shell, browser, sub-agents
OWNER	one person; British English
SPAN	24 Sep 15:38 – 25 Sep 17:17 EEST · 25 h 39 min
SOURCES	git (4 commits) · code at ddb01cc · session log · vaults

 River Portal DEVELOPMENT CARD 001	
MODEL	Claude Opus 5.5 claude-opus-5-5
DRAWN BY	the model, observing itself
CHECKED BY	the product owner
SOURCE	reflections/sdlc/v1
DATE · REV.	2026-09-25 · A
SHEET · SCALE	1 of 1 · 1 : 1, human